

→ When we use the Adjusted Present Value method, the answer that we get is different.

17th July 2026

Lecture-10

Adjusted present value Method

$$\text{Terminal unlevered value} = \frac{\text{FCF}_n (1+r)^{\text{risk free rate}}}{r-g}$$

$$\text{Terminal value Tax shield} = \frac{\text{Interest tax shield}_n (1+g)}{k_d - g}$$

$$\text{Unlevered Value} = \text{npv}(\rho, \text{First 4, last} + \text{Terminal value})$$

Mistake :- Using MM equation, based on the assumption that $g=0$

→ whenever

Mistake :- When you bring growth into picture, riskiness of the tax shield changes.

If the business is growing then few equations would change

Discount tax shield at ρ not k_d

$$\beta_L = \beta_U \left(1 + \frac{(1-t) D/E}{\rho} \right) - \beta_D \frac{D}{E} (1-t)$$

When $g=0$

→ Remove this

$$k_e = \rho + (\rho - k_d) \frac{D}{E} (1-t)$$

When D/E is kept constant

$$K_e = \rho + (\rho - K_d) \times D/E \quad [\text{If you keep } D/E \text{ constant}]$$

K_e changes

$(1-t)$ removed

~~WACC changes~~

~~$(1-t)$ removed~~

PV of tax shield uses ρ

and terminal value uses ρ

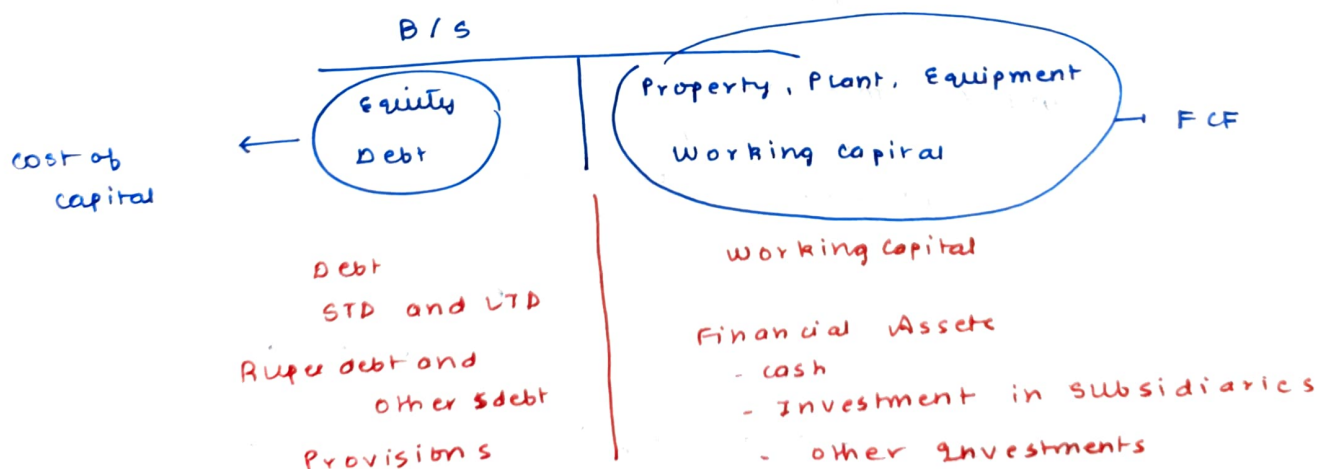
Example:

Year	0	1	2	3	4	5
Debt	100	80	60	40	20	0

cost of debt

B Cost of Capital

A simple valuation setting



Titan's Liabilities

→ Which numbers do I take while calculating debt.

Market Capital = No. of shares $\times$ Share price.

• Which figures to be included in debt

• Which figures to be excluded in debt.

→ When we ~~underst~~ get the Enterprise value, what number should we subtract to get equity value.

Four technical points about WACC

1. • Pre tax cost of debt is not the same thing as the coupon rate of bond → (YTM)
2. • Cost of equity is not the same as dividend yield.
↳ (dividend yield + capital gains)
3. • The weights in the computation should be based on MV and not BV
4. → The discount rate must match the riskiness of the cash flow.

⊗ All these 4 questions are present in the Book

1. It does not matter when $g = 0$

WACC comes out differently

✓ ↘
Ent Value ↑ Ent Value ↓

Problem comes when $g \neq 0$

Q6 The company gets a project with IRR = 17%

→ Should the company invest

↳ No.

How to tell that 17.57% is correct 16.87%.

Method I → What happens when the company invests in a project with IRR = 17%.

Initial investment 500 Mn and IRR 17% is available for the company. Should the company invest in this project?

↳ If they invest? → What would be the new Equity Value?

$$\text{Existing D/E ratio} \quad \frac{100}{315} \quad \frac{83.33}{315}$$

$$\frac{100}{415} \times 500 = \Delta D \quad \frac{315}{415} \times 500 = \Delta E$$

$$\text{IRR} = 17\% ; \text{Coupon} = 10\% ; K_d = 12\%$$

$$500 \text{M} \rightarrow \text{FCF} + 85 (17\%)$$

$$\text{EBIT} = \frac{85}{0.7}$$

$$= 121$$

$$\boxed{\text{EBIT}(1-T) = \text{FCF}}$$

NPV of this investment is ~~zero~~ negative.

$$\text{New EBIT} \rightarrow 121 + 100 = 221$$

$$315 + \Delta E =$$

- Why do we use market value based weights?

Book value share of Titan ₹ 230.19

Trading around ₹ 4,629

Titan's cost of equity is (12%)

↑
expected on this number.

→ Why should X bother about market value as promoter of the company.

↳ selling will happen at MV, hence return should be generated on MV.

Titan's Liabilities

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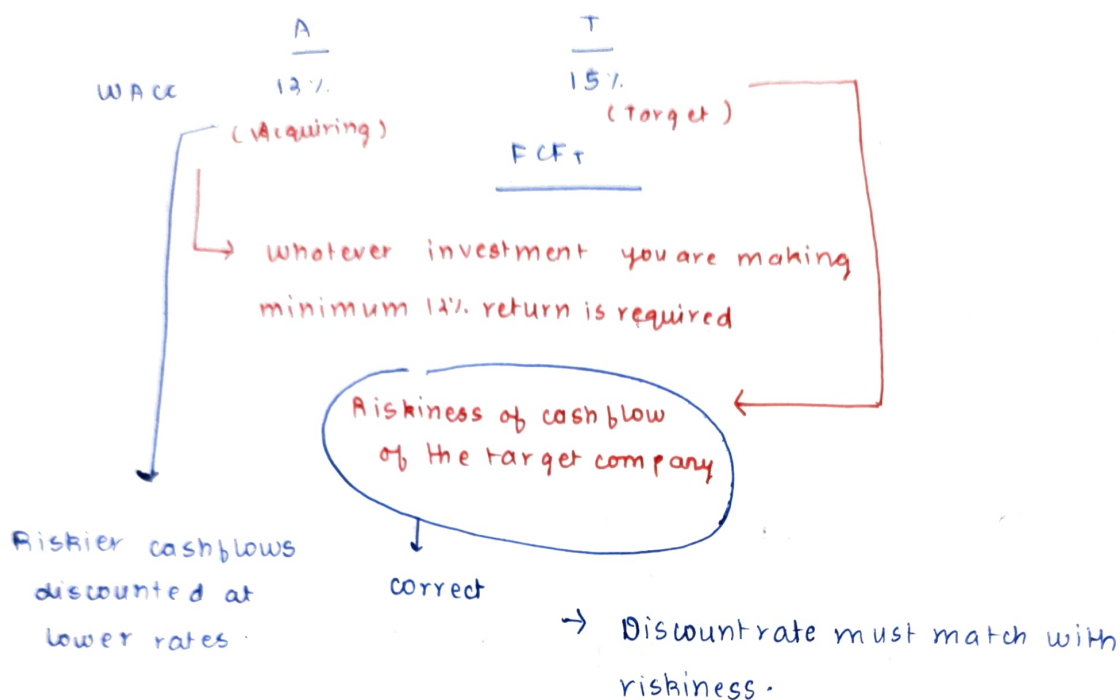
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Match riskiness to cashflows



Q: WACC 12%. Y-see 7%. (Investment)

→ What would ~~you~~ happen to your share price.

↓

Stay constant

↓

Zero NPV investment

Equity cash flow discounted at k_E

FCF discounted at WACC.

Match the Type

→ Nominal CF discounted at nominal cost of capital

→ Rupee rate CF discounted at Rupee cost of capital.

Take care of inflation.

→ Cost of Capital of company → Nominal cost of capital.

→ Would be ~~real~~ nominal

r_f → Nominal Rate.

$r_m - r_f$

Δ Revenue should have Δ P and Δ B.

→ Explicitly factored in inflation → Nominal Cash flow.

SBI = 9%.

EXIM = 3%.

↓

Bank in China

} → How would you calculate K_{cc}.

* convert Emirates Airlines cashflows to USD.

USD equivalent cost of capital.

22nd July 2026

Lecture-11

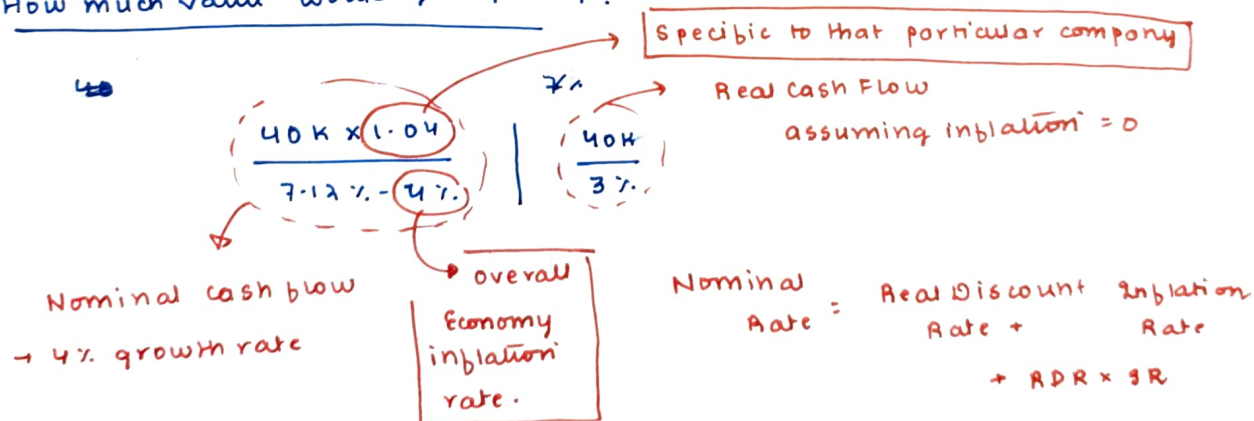
8.7 → Which cashflow to use Real or Nominal?

- Assume ASL Ltd. sells 1000 units of JLTs each year
- The unit selling price is ₹100 and the unit cost of production is ₹60.
- Assume that real discount rate is 3% and inflation rate 4%.

→ Each year how much cashflow are you expecting.

CF $40 \times 1000 = 40,000$ (Regular CF)

↳ How much value would you put up?



→ Assume that cost/sales ratio is constant.
Same inflation affecting Revenue and Cost of Production.

→ Inflation rate in numerator and inflation rate in denominator are never same.

Inflation is very high then it can be ignored.
Use Real CF.

Interest Rate Parity

- The exchange rate b/w \$ and ₹ is ₹ 80 per \$
- Interest rate in US 5% p.a
- Interest rate in India 7% p.a

→ If you have \$ 100, where will you invest?

$$\frac{80 \times 1.07}{1.05}$$

→ If expected rate is not this arbitrage will happen.

↳ Expected Forward price.

→ How much value the American investor put?

- US real discount rate = 3%.
- Inflation rate in US 1%.

Rs. Cash Flow

$$\frac{40,000 \times 1.04}{81.52} = \frac{CF_1}{81.52} = \$ 510.304$$

What would be the discount rate.

$$= \frac{\$ 510.304}{4.03\% - 1\%} \times$$

$$\downarrow$$
$$\left(\frac{\$ 510.304}{4.03\% - 1\%} \times 80 \right) \rightarrow \text{INR valuation}$$

→ When you are valuing a company and exchange rate is a problem.

value it using currency of your choice

↓
Main assumption: Interest Rate parity theorem holds.

If inflation is the problem, ignore inflation.

If exchange rate is the problem, ignore the inflation.

→ Regression of β of Infosys on the IT ~~index~~ index, but benchmark is the whole market.

→ Mistake 2 : → Regress β on S&P 500 not Nifty 500

→ which number would you use to find the cost of debt.

6% US borrowing

11% INR borrowing

↓

what would be K_d

6%	11%
US	INR
10,000	20,000

Estimating Cost of Debt

• The Pecking order

↳ what return market is expecting today

• Find the YTM of the bond (Pecking order 1)

↳ Yield is the best measure.

• Ask someone who knows (Pecking order 2)

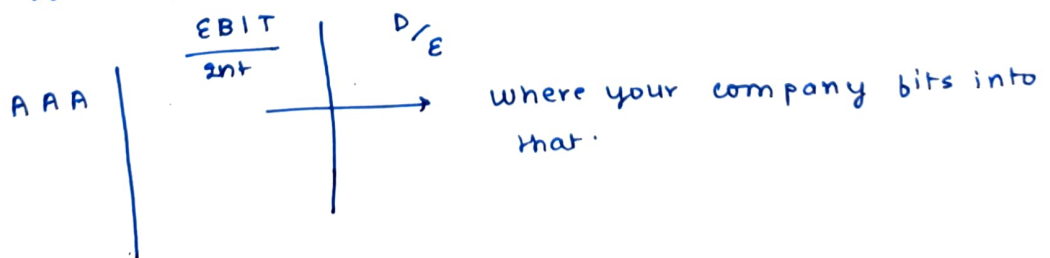
↳ someone working in Fixed Income.

• Can use Bloomberg.

• Use the credit Rating of the bonds issued by the company.
(Pecking order 3)

↳ Rating → SEBI → last 2-3 months → what coupon rate are they offering.

• Rate the bonds yourself. (Pecking order 4)



For AAA rating, what spread you would be using.

→ Rating from both Crisil and Kare.

* Market always follows the lowest rating.

Finance team and Marketing team should not talk to each other.

- 83 of the companies (BSE 500) have borrowed directly from the public as on FY 2023.

↳ Banking & Finance companies (80 companies).

- 146 companies have borrowed from the Bank.

- Rest companies have no debt.

Credit Rating and Yield Spread

- Data on corporate yield spread not available from any source.

- FIMMDA shares some data with the general public.

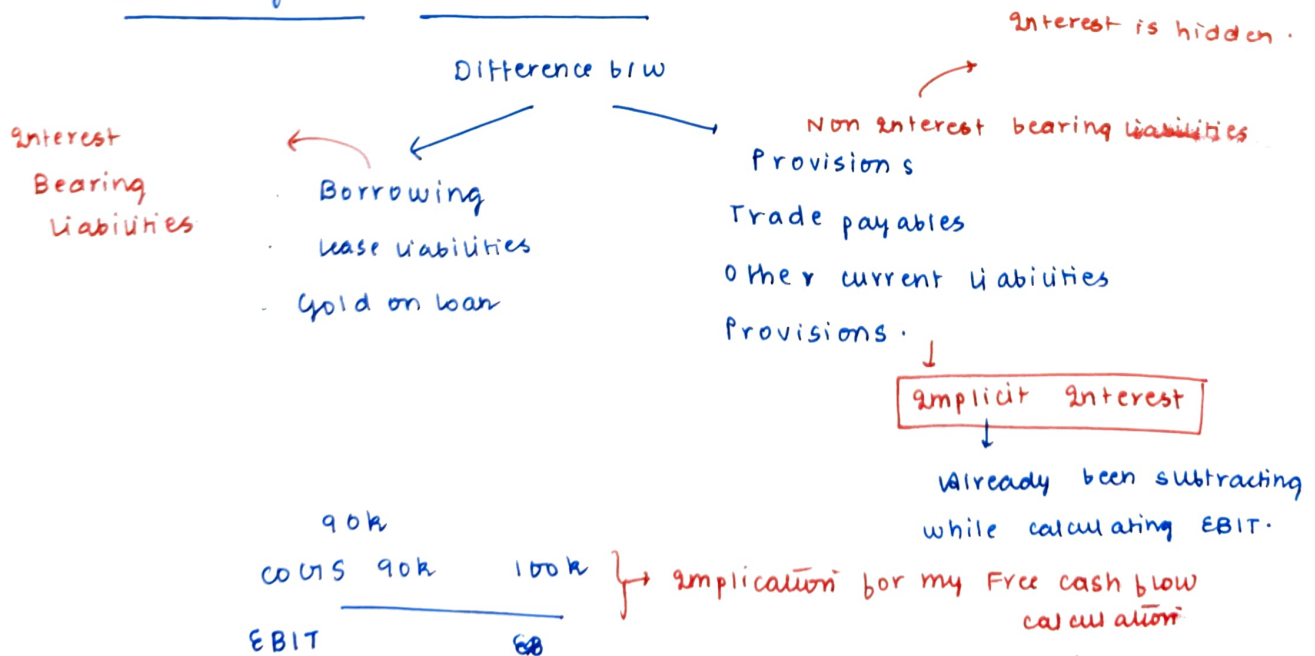
- Most bonds issued by SEBI are AAA rating.

- * Estimating cost of debt from coupon of a recently issued bond.

↓
↳ PDF File.

Cost of Debt from Yield.

Estimating Cost of Debt of Titan



How do I find my cost of debt.

→ XYZ limited has issued two types of debt at 12%.

EBIT 100M
int (40)

$$E = 210M$$

$$EV = 510M$$

WACC =

PBT 60M
Tax (18)
PAT 42M

$$\frac{210}{510} \times 20\% + \frac{100}{510} \times 8.4\% + \frac{200}{510} \times 9.8\%$$

$$= 13.72\%$$

$$FCF = 70$$

$$EV = \frac{70}{13.72\%} = 510$$

* Any liability you are not sure about cost of debt, ignore those liability.

Ignore Debt 1 [Company is paying both the interests]

$$\frac{210}{510} \times 20\% + \frac{200}{510} \times 9.8\% = 15.02\%$$

$$FCF = 88(1 - 30\%) = 61.6$$

$$EV = \frac{61.6}{15.02\%} = 410$$

EBIT 100M
int (12 + 28)

New EBIT 88
int (28)

PBT 60
tax (18)

* When liability is creating a problem, treat that liability as operating liability.

There is something called the Equity value, but not the WACC, not the K_d .

FCF and WACC cannot be independently determined.

FCFE method, special case FCF method where all liabilities are operating liabilities.

Decide to ignore the liability for which numbers are hard to get.

Just ignore the current liabilities.

Consistency Issues

2b D_1 is a source of capital - $WACC + f(D_1)$...

Donot subtract interest on D_1 while computing FCF

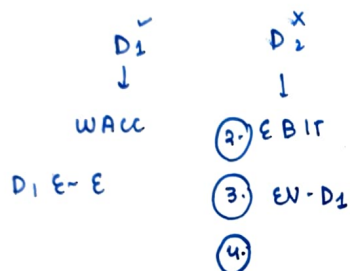
24th July 2026

Lecture-12

2b D_1 is a source of capital

Donot subtract interest on D_1 while computing FCF

Subtract D_1 from Enterprise Value, while computing Equity



Adjustments

→ 2b D_0 is not a source of capital

Do subtract interest on D_0 while computing FCF

Add increase in D_0 , while computing FCF

$FCFE = (+ \Delta D) \rightarrow$ change in Debt which was added, was the debt we didnot consider.

Increase in liability which was not included, needs to be added back.

Consistency Issues → 4 issues → 4 adjustments.

PV (Lease Rentals)

→ cost of capital in any retail company?

↳ Should PV (LR) be included or not $\left\{ \begin{array}{l} \checkmark D_1 \end{array} \right.$

→ when calculating EBIT what adjustment should be made.

- operating lease would include an interest and depreciation component.
then it becomes D_2 .

If something is becoming very complicated then consider it as D_0 , but remember the adjustments required.

Q: $D = 1000 \text{ M @ } 8\%$
 $CMS = 200 \text{ M}$
 $K_e = 13\%$ } while calculating cost of capital, what would I do with cash.

Gross Debt	Net Debt
D, E (ignore cash)	$D - C, E$
$GD = 1000$ $ND = 800$	* cash is being treated as Non-operating asset
→ Which method is correct?	

$WACC \rightarrow f(D, E, \dots)$

EBIT → Interest on Debt not subtracted

→ Interest on C will be added to

3 sources of funding capital

$D, -C, E$ → Negative Debt → I already have that money.
G.D Method ✓ x ✓ → Sources of Capital.

In the gross debt method we consider D, E and ignore C .

$$3.) \text{ Equity} = EV - D = E - (-C)$$

What happens in the Net-Debt Method?

1.) $WACC = f(D - C, E, \dots)$

2.) EBIT → Do not subtract interest expense, do not add interest income.

3.) Equity: $EV = (D - C)$

→ If the company has surplus cash, it would misuse that surplus cash.

* Most people who use net debt method make this mistake.

What if cash is present?

$$\frac{ND}{ND+E} \times (K_d)(1-t)$$

Cost of Foreign Currency Loans

Assume company

EBIT = 100 (Does not include interest income)

Debt = 100 (@ 10%)

Cash = 50 (@ 7%)

Tax Rate = 30%

Kc = 20%

PBT = 93.5

100 - 10 + 3.5

Net Income = 63.5

Equity = 327.25

$$\frac{6.5\%}{50} = 13\%$$

Effective cost of Borrowing.

What if cash is present?

~~FCF = 70~~ . Cash is non-operating

EBIT ≠ Net Debt method

EBIT = 100

FCF = 70

[Considering everything]

Gross Debt Method

EBIT = 103.5

FCF = 103.5 × 0.7

[D, E]

Ignoring -c

↓

- (-3.5)

= 3.5

FCF = 70

EV = 327.25 (= 327.25 + 100 - 50)

WACC = 18.55%

Effective cost of Debt is 13%

$$= \left(\frac{100}{327.25} \times 10\% \times (1-30\%) - \frac{50}{327.25} \times (7\%) \times (1-30\%) \right)$$

$$= \frac{327.5}{327.5} \times 20\%$$

$$= \frac{50}{377.75} \times 13\% \times (1 - 30\%) \rightarrow \text{identical numbers.}$$

$$EV(\text{check}) = 70 / 18.55\% = 377.75$$

10 different cost of capital but always 1 cost of equity number.

→ Next level complication of introduce growth rate?

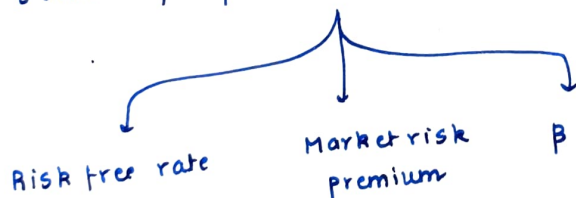
Cost of Equity (Ke) → is defined from the investors perspective.

CAPM: → outdated; now Fama-French 5 factor model is used.
Things started changing in 1992

→ All the 10 portfolios gave similar results, difference was not significant.

High β and low β gave similar returns.

$$E(r_i) = r_f + \beta (E(r_m) - r_f) \rightarrow \text{only asset pricing model which can be proved.}$$



→ Should not use when valuing a private company.

Alternatively, use Multi-factor Models.

Arbitrage Pricing Theory (You need to estimate the factor premiums).

Risk-free rate: Yield curve is not flat.

→ is not one number.

Risk-free rate tenor must match cash flow tenor.

• Longest ~~of debt~~ term rate.

Maximum liquidity at 1 Year or 10 Year

CDS: → Indian Government Bonds → Yield curve.

McKinsey's argument: → Country risk premium, something which can be diversified.

Rf: → Yield curve is not flat, how to find Rf.

Year	1	2	3
CF	100	100	200

Year	1	2	3
Spot Rate	5%.	6%.	7%.

$$\text{Value} = \frac{100}{1.05} + \frac{100}{1.06^2} + \frac{200}{1.07^3} = 347.50$$

$$347.50 = \frac{100}{(1+r)} + \frac{100}{(1+r)^2} + \frac{200}{(1+r)^3}$$

$$r = 6.52\%$$

↳ Yield to Maturity.

→ If the final Years cash flow is a very large number then this affects your yield.

Market - Risk Premium

$$\frac{r_i}{\beta} = \frac{r_m}{\beta} - r_f$$

average of this.

$$r_m - r_f$$

1984 - 2026

sensex.

→ If you are taking large data.

Taking large amount of historical data.

Go to ABI website: rm - rf Historical Data.

• Do not calculate $rm - rf$ on your own.

29th July 2026

Lecture - 13

Estimating Beta

$$\beta = \frac{\text{Covariance}(R_f, R_m)}{\text{Variance}(R_m)} = \frac{\sigma_{i,m}}{\sigma_m^2}$$

• How to compute returns

$$\frac{P_t}{P_{t-1}} \rightarrow \text{Prowess}$$

$\ln\left(\frac{P_t}{P_{t-1}}\right) \rightarrow$ Return always used in model assuming normal distribution.

→ Quadratic

→ CAPM assumes normal distribution.

• either utility is quadratic



→ skewness and kurtosis would not affect the CAPM function.

→ $\ln\left(\frac{P_t}{P_{t-1}}\right) \rightarrow$ limitation does not work in a portfolio

Yahoo Finance would give closing price or adjusted price.

1. Dividends
2. Bonus
3. Split
4. Rights issued.

→ These 3 adjustments must be made.

PCR 100
Issue: 80 (1:1) } 1 Earlier share¹ worth ₹100, now becomes 2 Shares of ₹180.

$E(P_{XR}) : 90 \rightarrow$ call option.

$$\frac{100}{90} - 1 = 1.1 \rightarrow \text{Bonus Issue.}$$

180 with you

→ 1.9 shares can be bought 1.8

with rights: 2

company has given you a bonus of 0.2
for every 1.8

$$\frac{0.2}{1.8} = 1/9.$$

$$\frac{P_{CA}}{P \times A} - 1$$

→ Bonus Ratio

→ same adjustments for bonus.

Estimating Beta

→ What should be the estimation period?

10 year > 5 year > 1 year

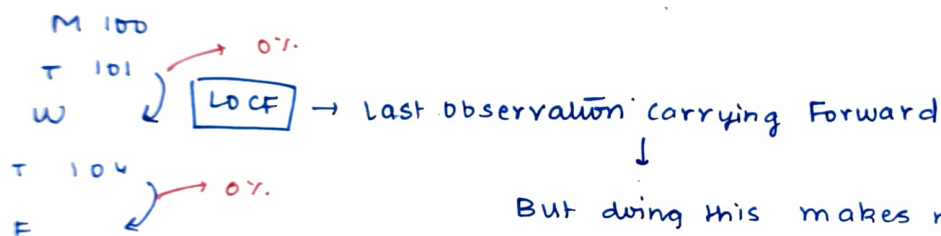
↳ During your estimation, the risk of the company has changed.

→ That period, which looks similar to what the company is doing.

↳ Hence 10 year data not ϕ better.

Rule of Thumb: → 1 Year data for a company trading daily is enough.

→ How frequently should we compute returns?



$$\text{Covariance}(S, M) = \rho_{S, M} \times \sigma_S \times \sigma_M$$

standard deviation of stock

Standard Deviation of market

β under

· When stock is not traded, β would be 1, this does not mean that stock is less risky. Measurement error.

β of a relatively illiquid stock would be undercalculated.

· Better to use weekly returns.

→ When you are using monthly returns, you have to use 5 year period.

→ operating and financing risk of the company should not have changed in the period.

Value Investing → Bruce Greenwald.

Bloomberg terminal

Raw $\beta \rightarrow 1.717$
Adjusted $\beta \rightarrow 1.478$

β increases from 1.348 to 1.717 when monthly returns from last 60 days are used.

$$\beta_{t+1} = \frac{1}{3} + \frac{2}{3} \beta_t \rightarrow \text{Merryl and Lynch}$$

↳ study in 1980

will have a tendency to move towards 1.

Low Beta (MCF)

Unusual value of $\beta \rightarrow$ Data Problem

$$\left(\frac{\partial EV}{\partial WACC} \right), \quad \frac{\partial EV}{\partial M}, \quad \left(\frac{\partial EV}{\partial g} \right)$$

Highly dominating

↳ Not that dominating

→ Hence while valuing you should not spend much time estimating what growth would be.

Cost of Warrants → special type of call options

↳ ESOPS → call option given by the company.

$$\text{Warrant} = \frac{\text{Call}}{1 + p}$$

$$p (\text{dilution factor}) = \frac{nw}{ns}$$

First estimate the β of call option.

$$\beta_{\text{call}} = N(d_1) \cdot S_{\text{EW}} \cdot \beta_E \quad \text{↳ } \beta \text{ of equity}$$

$$c = S N(d_1) - K e^{-rt}$$

* Warrants → Preferential allotment of stock, at a price lower than market price.

Cash Flow and Valuation

$$\text{Value} = \frac{\text{FCF}_1}{(1 + \text{WACC})} + \frac{\text{FCF}_2}{(1 + \text{WACC})^2} + \dots = \sum_{t=1}^{\infty} \frac{\text{FCF}_t}{(1 + \text{WACC})^t}$$

Value of company = Present value of cash flows during the explicit forecasting period + Present value of cash flows beyond the explicit forecasting period (terminal value).

↳ Although Benchmark is 5 years, but do use your judgement.

NYSE listed company full report is available.

A few starting problems

More than 50% of a company consider it as a subsidiary.

P Limited

P Limited's equity ownership

Net Assets	£ 500 million
Net Income	£ 100 million
Market Cap	£ 1000 million

$$1000 \text{ M} = P + 80\% \text{ of } S + 20\% \text{ of } M$$

$$680 \quad 240 + 80$$

$$(300) \quad (40)$$

Get 680, from stand alone financials.

Consolidated

$$980 = 680 + 300$$

$$+ 80$$

$$- 20\% \text{ of } 300$$

If you are consistent it does not matter

↳ Is it easy to be consistent

↳ NO.

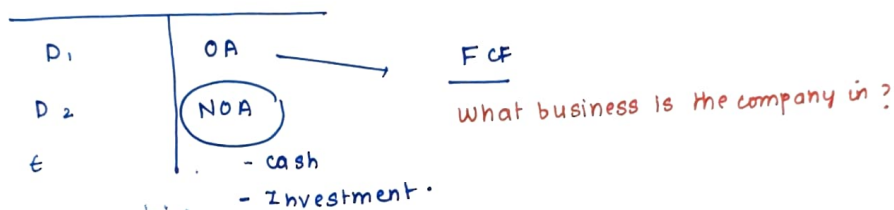
In Indian context always better to look at consolidated B/S. standalone.

→ Market value of Investments

standalone financials and then the adjustments.

Maintaining Consistency in Cash Flow

↳ in Assets & liabilities & cash flows & WACC.



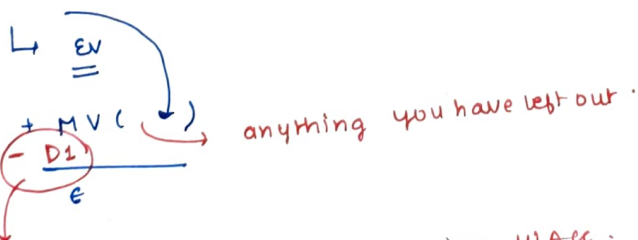
$$WACC = f(D_1, \epsilon)$$

Valuation of Bajaj Auto

- Am I valuing a two wheeler company.
- Am I valuing a mutual fund company

FCF should come from operating assets.

FCF @ WACC



Number used while calculating WACC.

D₂ has been considered while calculating cash flow.

Orion AI from vestai.io

Final Project Report

Do not make any consistency error.

Maintaining Consistency in Assets & Liabilities & Cash Flow & WACC

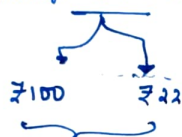
Understanding the different line items

- Line items may be different in service-oriented companies

Revenue

- Sales or Net sales?

- 0% rooms used to charge 22% commission from the hotel partners



Revenue for cost of goods sold would ensure that FCF is correct.

→ This number matters, when we are using revenue based multiplier. Should be consistent across all peers.

* May need to restate previous year's figures

- We need a base number to forecast future sales.

e.g. Walmart international revenue fell from \$131.4 Bn in 2021 FY \$101 Bn.

30% of the time on getting the revenue number correct.

Other operating Income

Projection should be based on these numbers.

↳ Non-Recurring

- Is it part of the main business activity?

- Is this recurring income?

Examples (operating or non-operating)

- Income from credit card operations retail firm + operating

↓
No corresponding asset generating this cash flow.

↳ Recurring.

If you are using Gross Debt to calculate WACC, we treat cash as operating income.

→ Drug retailer gets revenue from selling containers.

↳ If I exclude this 10% income, there is no asset in the balance sheet, through which I can adjust.

It answer to this question is 'NO'
Must Include.

→ gain on sale assets should be included?

↳ Depreciation in B/S is 10, but actual depreciation only 5 (Economic value)

* What should be done? To balance added back.

$$\begin{array}{r} PV(CF) \\ MV(\text{Brand}) \\ - D \\ \hline E \end{array}$$

→ This number is already included in your FCF calculation, Adding that back again only creates a problem of double counting.

→ Should we add this number.

Real Estate not in your B/S?

$$\begin{array}{r} PV(CF) \\ A(MV-BV) \\ - D \\ \hline E \end{array}$$

→ Shalimar Pens, lot of real estate which they purchased at a very low price.

* If you have surplus land, only then add back, if your factory is located on this land do not add!!

Total operating Income = Net sales + other operating income.

OPEX line items

- Raw Materials
- Wages and salaries
- other manufacturing expenses.
- selling and Distribution expenses
- Administrative expenses.
- other expenses (put all smaller items here).
- Depreciation.

* compute the % of total sales (or fixed assets) over the last 3-5 years.

→ Compute EBITDA and EBITDA margin

Forecasting Depreciation

Net operating Income = Total operating Income - Total operating Expenses.

Net operating Profit Adjusted for Tax + Depreciation = Gross cash flow

Not adjusted anywhere else.

* So many non-cash expenses, but only depreciation is added back. Because depreciation and amortization have not been adjusted anywhere.

added back alone
→ Key word

Estimating Gross Investment

- computing capex from financial statements

* under US GAAP, depreciation is not a separate line item.

Best place to find depreciation CF statement

$$\text{Capex} = \Delta \text{NFA} + \text{Dep'n} + (-\text{M} + \text{L})$$

→ Compute increase in operating working capital

• Operating and Non-operating cash.

$$(\text{CA}) - \text{CL}$$

$$(\text{CA} - \text{cash}) - (\text{CL} - \text{IBL})$$

* What other adjustments.

operating working capital.

WACC

Net debt

EBIT → do not add interest on cash.

Net Debt { STD = Source of capital, hence ^{not} source of cash.
LTD

$$\frac{\text{EV} - \text{ND}}{\text{E}}$$

$$\text{Operating Working Capital} = (CA - \text{cash}) - (LI - IBL)$$

When calculating Net Debt, STD is treated as a source of cash, hence cannot be a source of capital.

* For many profit making companies WC is negative

→ Working Capital of Bharti Airtel Ltd.

* What is the problem with negative working capital?

$$GCF, CFI = FCF$$

capex

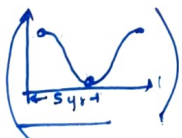
$$\Delta WC \rightarrow f(\text{Revenue})$$

Q: When does WC become a source of cash?

→ Rev ↑ → WC becomes more -ve, hence $FCF \uparrow$

* Making WC a source of cash.

→ Would you consider the privillages of the company to continue?

Steel company Capex is lumpy  → Forecasting about next round of capex.

→ What is the capacity utilization and when would they be utilizing their maximum capacity.

$$FCF = NOPAT - \text{Net Investment}$$

* can you calculate this number from C/F → NO
↳ This is adjusted tax and not actual tax.

* Free Cash Flow → -100 Cr
→ spend more than you are making.

Free cash flow would fluctuate a lot.

Valuing a company under financial distress → High learning value.

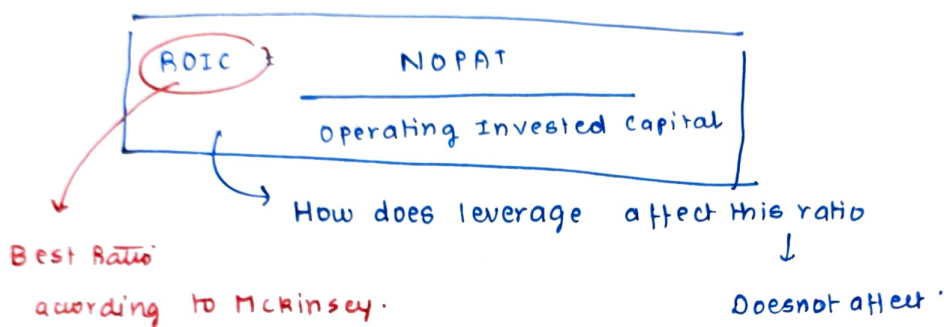
→ For which companies B/S becomes completely irrelevant?

Operating Fixed Assets + Operating wkg. Capital + other operating assets.

Which item is being ~~in~~ excluded

↳ capital wkg.

↳ Factory construction happening.



ROE → Affected by leverage.

$$\text{ROE} > K_e$$

ROIC can be compared with WACC

$$\text{EVA} = \text{Operating Invested Capital} \times (\text{ROIC} - \text{WACC})$$

↳ Economic Profit

7th August 2026

Lecture-15

→

- 1.) Companies she tracks → Inbo Public | Inbo Private
- 2.) Buy side clients
- 3.) Internal.

Laura Martin talks to these 3 stakeholders.

\$ largely recommendations about Buy.

Hold recommendations stock is the same

12. sell recommendations to please companies she tracks

* If an analyst would give buy recommendation → stock market would have no effect.

→ But, when sell recommendation the market reacts.

→ sell recommendation people do take things personally

Buy side more unbiased as compared to sell side.

→ In Buy side you get paid to outperform your index, sell side more about communicating the results of your analysis.

Quid Pro Quo Buy side clients?

- ↳ Information
- ↳ credibility → Institutional Rankings.

HBA Star Analyst

↳ Buy Recommendation

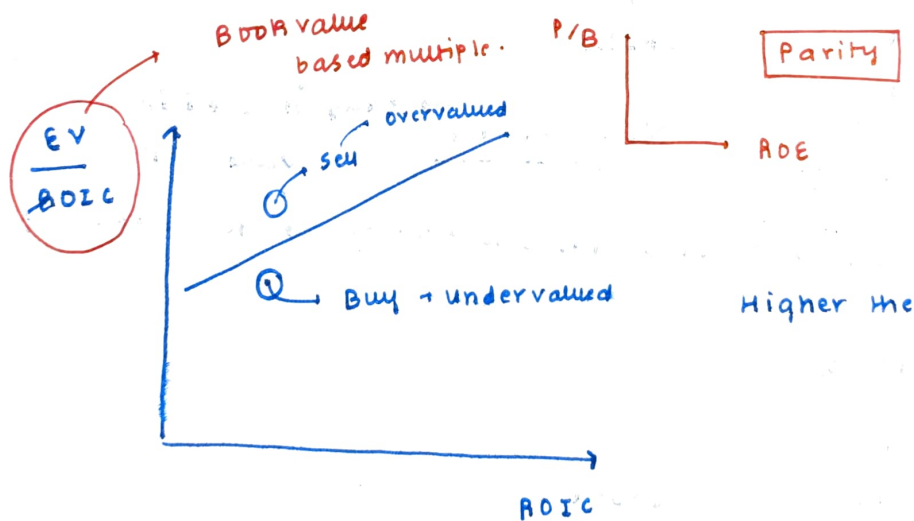
Internal • Trading Desk

• Sales Force

• Investment Banker

• M & A

• Funding



$$R_1 = (ROE - K_e) \times N \cdot W$$

$$EVA = (ROIC - WACC) \times OIC \rightarrow \boxed{EV = OIC + PV(EVA)}$$

$$EV = NW + PV(R_1)$$

→ Laura Martin is transferring this.

- * As if all companies within the industry would be equally profitable.
[Limitation of multiplier method]

 overvalued but recommendation to buy

ROIC of COX has to increase 80 basis points
Not sufficiently move below the valuation line.

COX communication 3 businesses

Discounted cashflow method for COX communication

→ capex going down and then becoming constant.

→ Is it a problem, when capex keeps on decreasing?

2008 Terminal year

$\frac{FCF_{2008} (1+g)}{WACC - g}$ FCF growing at the constant rate.

Capex 261M

Dep'n 515M

Minimum = 0

$WACC - g$
(9.3%)

NI = -254M → okay in explicit valuation.

→ By how much has she overvalued the company?

$g = 0\%$

$$\frac{254}{9.3} \bigg/ 565.2$$

$(1.093)^{10}$

$$\frac{1804 (1+g)}{9.3\% - g} = 38,454$$

After adding 1504.

→ circularity problem.

$$\frac{1804 - 254}{9.3\%} = 16,000\%$$

$$\frac{1804(1+g)}{9.3 - g} = 38454$$

→ $g = 4.4\%$ → Wrong Number.

NI when $g = 4.4\%$.

$$\frac{1804 - 254 - NI}{9.3\% - g} =$$

Equivalence of various terminal multiples and values.

$$g = \text{Retention Ratio} \times \text{RDE}$$

How much investment

$$g = \text{ROIC} \times \left(\frac{\text{Net Investment}}{\text{Profit}} \right) \quad \text{NOPAT}$$

$$= \text{ROIC} \times \left(\frac{\text{Net Investment}}{\text{NOPAT}} \right)$$

Analogous to Retention Ratio.

$$g = \frac{\text{NOPAT}}{\text{OIC}} \times \frac{\text{NI}}{\text{NOPAT}} = \frac{\text{NI}}{\text{OIC}}$$

Investment Rate.

$$\begin{array}{r} 459 \\ + 254 \\ \hline 713.04 \end{array}$$

Amount by which she has overstated.

$$\text{OIC}_{1999} + \text{NI}_{2000} + \dots$$

Capex - Dep

$$\frac{4.4}{100} = \frac{\text{NI}}{\text{OIC}_{2007}}$$

$$\text{OIC}_{1999} = 12,136$$

$$\text{OIC}_{2007} = 10,441$$

$$\text{NI}_{2007} \rightarrow 1695$$

$$12,136 - 1695$$

$$= 10,441$$

$$\text{NI} = 4.4\% (10,441) = 459.404$$

Actual net investment